



Johannes-Brahms-Schule Pinneberg

Physics Major Ea

How Chaotic is Chaos?

*A Numerical Analysis of Double Pendulum
Dynamics.*

Instructor:
Steffan Stoppel

Submitted By:
Alireza Khodabakhshian
Gmail: alirezakhodabakhshian14@gmail.com

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Preface

I would like to express my sincere gratitude to my parents for giving me the opportunity to pursue this work and for supporting me throughout my life in countless ways. I am especially grateful to my mother, my father, and my brother for encouraging me while I was preparing and conducting this research. Without their support, I would not have been able to complete this paper.

I would also like to thank my teacher, Mr. Steffan Stoppel, who guided me as my instructor and helped me many times during the writing of this paper, especially with the structure, formatting, and style of the work. His advice and support were very valuable to me.

From an early age, I developed a deep interest in mathematics, and when I encountered physics in the seventh grade, that interest became a true passion. In particular, I have always been drawn to theoretical physics. As Paul Dirac famously suggested, the beauty of a physical law and the elegance of its mathematical form are of great importance. This idea has always inspired me and guided my curiosity.

Although this is an experimental research paper, it also includes many beautiful theories that allowed me to explore, reflect, and learn about advanced physical concepts. In preparing this project, I tried to present the necessary theory as clearly as possible so that the experiment could be understood in a meaningful way. I hope that the reader will follow both the theoretical background and the experimental part with interest.

I have always enjoyed challenges, and that is one of the reasons I chose such an ambitious project. Building the double pendulum alone required more than €100, and it demanded patience, effort, and careful work. Through this research, I wanted not only to complete a project, but also to deepen my understanding of physics and contribute, in my own small way, to the subject I admire so much.

Finally, I hope to honor the legacy of great physicists such as Richard Feynman, Nikola Tesla, Albert Einstein, Erwin Schrödinger, Paul Dirac, and others who devoted their lives to science. My hope is that this paper will reflect the curiosity, beauty, and wonder that first made me fall in love with physics.

Introduction

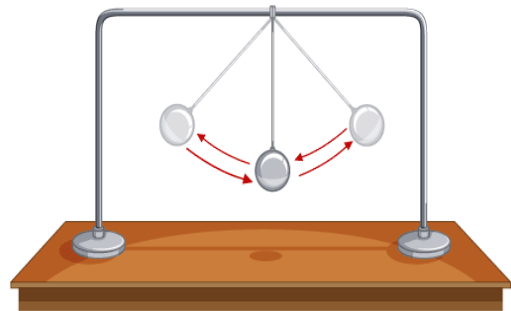
This research aims to illustrate how chaos arises by numerically analysing a double pendulum. Using the double pendulum as a concrete model, I develop the Lagrangian equations of motion, explore the system's nonlinear dynamics, and relate those dynamics to core ideas in chaos theory. The project also provides a practical toolkit: a clear methodology for numerical integration implemented in Python, a description of what a precise experiment would look like, and an investigation of the practical and theoretical limits of predictability. More than an account of striking behaviour, this work seeks to clarify what physicists mean by “chaos” and to propose quantitative ways to measure predictability in real experiments. The material is written for learners who want to bridge theory and practice — including advanced high-school students and first-year undergraduates — and has been organised to be both accessible and formally precise.

Basic Physical Knowledge

The simple pendulum basics

What is a simple Pendulum?

A pendulum consists of a mass suspended from a fixed pivot, allowing it to swing freely. When the pendulum is displaced from its resting, or equilibrium, position and released, gravity accelerates the mass back toward equilibrium. This gravitational force acts as a restoring force, causing the pendulum to oscillate about the equilibrium point, swinging back and forth on either side (Taylor, 2005; Goldstein, Poole, & Safko, 2002).



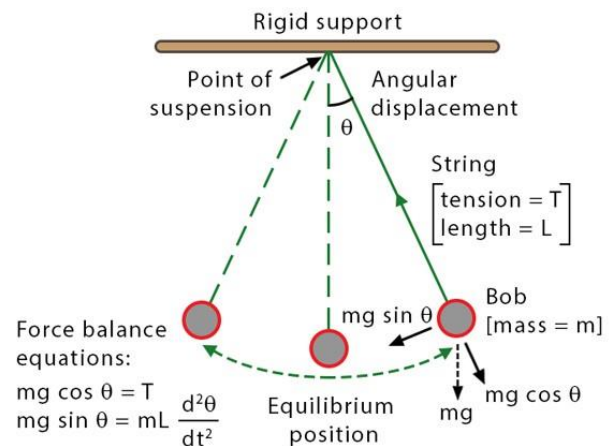
Before the analysis, note that :

- ☐ The string is massless and does not stretch.
- ☐ The mass (bob) is a point mass.
- ☐ Air resistance and friction at the pivot are negligible.

Important physical terms :

The pendulum is suspended by a string of length L , and the gravitational force ($F_g = mg$) acts directly on the mass, pulling it downward. The string exerts a tension force on the mass, which balances the vertical component of the gravitational force. The horizontal component of gravity acts as the restoring force, pulling the mass back toward the equilibrium position. We

Simple Pendulum



define the angle θ as the angular displacement between the mass's current position and the equilibrium position. From this definition, we can write the force-balance equations. Because the mass remains constrained by the string, the tension force must balance the radial (vertical) component of gravity, which is $mg \cos\theta$. The tangential (horizontal) component of gravity provides the restoring force, defined as $F = mg \sin\theta$, is equal to mL times the second derivative θ of with respect to time (Taylor, 2005; Goldstein et al., 2002).

Additionally, the time required for the pendulum to complete one full oscillation and return to its original position is called the **period**, denoted by T . The angular velocity, ω , is defined as the time derivative of the angular displacement θ :

$$\omega = \frac{d\theta}{dt}$$

The reciprocal of the period T is called the frequency, denoted by f . The frequency in our case gives us the number of oscillations that happen per unit of time, commonly measured in Hertz (Hz).

For a simple pendulum, the period is independent of the mass and is determined by the length of the string and the strength of gravity:

$$T = 2\pi\left(\frac{L}{g}\right)^{\frac{1}{2}}$$

(The derivation and domain of validity are standard; see Taylor, 2005.)

Energy Transformation:

The motion involves a continuous exchange between **gravitational potential energy** ($U = mgh$) and **kinetic energy** ($k = \frac{1}{2}mv^2$), and the total mechanical Energy E remains constant:

$$E = \frac{1}{2}mv^2 + mgh$$

$$\dot{E} = 0$$

in the absence of non-conservative forces (Goldstein et al., 2002).

Note:

1. For small displacements (typically below 10–15° we apply the Small Angle Approximation, which allows the motion to be treated as Simple Harmonic Motion (SHM).
2. We will not derive these equations here, as they are standard results assumed to be familiar. Readers who are not comfortable with them are encouraged to consult a classical mechanics reference.

Advanced Mechanics

The Lagrangian & Equations of Motion

In this chapter we move from Newtonian single-pendulum analysis to a systematic derivation of the equations of motion for the planar double pendulum using the Lagrangian formalism. The Lagrangian approach uses generalized coordinates and energy functions to produce equations of motion compactly and without explicitly writing constrained vector force balances; it is therefore well suited to coupled, constrained systems such as the double pendulum. We derive the kinetic and potential energies, form the Lagrangian $\mathcal{L} = T - V$, apply the Euler–Lagrange equations, and produce the coupled second-order ODEs used for analysis and numerical simulation.

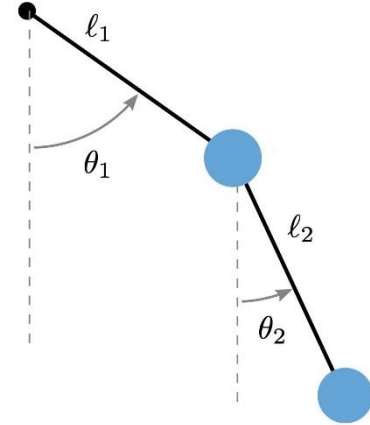
Notation & assumptions:

We denote masses m_2, m_1 , lengths l_1, l_2 , angles θ_1, θ_2 (measured from the downward vertical), and use SI units. We assume point masses, massless rigid rods, planar motion and no non-conservative forces unless stated otherwise.

1. Mechanical model and generalized coordinates:

We consider an idealized planar double pendulum consisting of two point masses m_1 and m_2 attached to two rigid, massless rods of lengths l_1 and l_2 . The first rod is fixed at a pivot, and the second rod is attached to the end of the first. Motion is confined to a vertical plane, and gravitational acceleration g acts downward. No friction, damping, or external driving forces are included.

We define the generalized coordinates θ_1 and θ_2 as the angles each rod makes with the downward vertical direction. The Cartesian coordinates of the two masses are therefore



$$x_1 = l_1 \sin \theta_1$$

$$y_1 = -l_1 \cos \theta_1$$

$$x_2 = l_1 \sin \theta_1 + l_2 \sin \theta_2$$

$$y_2 = -l_1 \cos \theta_1 - l_2 \cos \theta_2$$

2. Kinetic and potential energy; Lagrangian formulation:

Differentiating the position coordinates with respect to time yields the velocities $\dot{x}_1$ and $\dot{y}_2$

The total kinetic energy of the system is

$$\begin{aligned} T &= \frac{1}{2} m_1 (\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2) \\ &= \frac{1}{2} m_1 \dot{\theta}_1^2 l_1^2 + \frac{1}{2} m_2 (\dot{\theta}_1^2 l_1^2 + \dot{\theta}_2^2 l_2^2 + 2 l_1 \dot{\theta}_2 \dot{\theta}_1 l_2 \cos(\theta_1 - \theta_2)) \end{aligned}$$

The gravitational potential energy, taking the zero reference at the pivot, is

$$V = m_1 g y_1 + m_2 g y_2 = -(m_1 + m_2) g l_1 \cos \theta_1 - m_2 g l_2 \cos \theta_2$$

The Lagrangian of the system is defined as

$$\mathcal{L}(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2) = T - V$$

3.Euler–Lagrange equations and equations of motion:

The equations of motion are obtained from the Euler–Lagrange equations,

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_i} \right) - \frac{\partial \mathcal{L}}{\partial \theta_i} = 0 \quad i = 1, 2.$$

(Goldstein et al., 2002).

Applying these equations to the Lagrangian above and simplifying yields the coupled second-order differential equations:

$$\begin{aligned} (m_1 + m_2)l_1\ddot{\theta}_1 + m_2l_2\ddot{\theta}_2\cos(\theta_1 - \theta_2) + m_2l_2\dot{\theta}_2^2\sin(\theta_1 - \theta_2) + (m_1 + m_2)g\sin\theta_1 &= 0 \\ m_2l_2\ddot{\theta}_2 + m_2l_1\ddot{\theta}_1\cos(\theta_1 - \theta_2) - m_2l_1\dot{\theta}_1^2\sin(\theta_1 - \theta_2) + m_2g\sin\theta_2 &= 0 \end{aligned}$$

These equations fully describe the dynamics of the planar double pendulum and form the basis for the numerical analysis.

4. Explicit acceleration formulas for numerical integration

For numerical integration we use algebraically explicit expressions for the angular accelerations. Define the relative angle $\Delta = \theta_1 - \theta_2$. A common and robust form for $\ddot{\theta}_1, \ddot{\theta}_2$ (for point masses on massless rods) is:

$$\ddot{\theta}_1 = \frac{-g(2m_1+m_2)\sin\theta_1 - m_2g\sin(\theta_1-2\theta_2) - 2\sin\Delta m_2(l_2\dot{\theta}_2^2 + l_1\dot{\theta}_1^2\cos\Delta)}{l_1(2m_1+m_2-m_2\cos(2\Delta))}$$

$$\ddot{\theta}_2 = \frac{2\sin\Delta((m_1+m_2)l_1\dot{\theta}_1^2 + g(m_1+m_2)\cos\theta_1 + m_2l_2\dot{\theta}_2^2\cos\Delta)}{(l_2(2m_1+m_2-m_2\cos 2\Delta))}$$

This algebraic form is algebraically equivalent to the implicit pair from the variational derivation and is widely used in numerical implementations (Goldstein et al., 2002; Strogatz, 2018).

5. First-order state-space formulation

To integrate the system numerically, we convert the second-order equations into a first-order system by defining the state vector

$$\mathbf{y} = \begin{pmatrix} \theta_1 \\ \dot{\theta}_1 \\ \theta_2 \\ \dot{\theta}_2 \end{pmatrix}$$

The time evolution is then governed by

$$\dot{\mathbf{y}} = \begin{pmatrix} \dot{\theta}_1 \\ \ddot{\theta}_1(\mathbf{y}) \\ \dot{\theta}_2 \\ \ddot{\theta}_2(\mathbf{y}) \end{pmatrix}$$

Where $\ddot{\theta}_1$ and $\ddot{\theta}_2$ are given by the explicit formulas above (see Methods for solver details; Hairer, Lubich, & Wanner, 2006; Virtanen et al., 2020).

6. Total energy and conservation

The total mechanical energy of the system is

$$E = T + V$$

with T and V as defined in Section 2. In the absence of non-conservative forces, E is conserved. During numerical integration, monitoring the relative energy drift

$$\frac{|E(t) - E(0)|}{|E(0)|}$$

provides a quantitative measure of numerical accuracy and stability (Hairer et al., 2006).

Note: In the theoretical treatment we assume a conservative system. In the physical experiment, energy dissipation due to pivot friction and air resistance is present, which causes amplitude decay over time. The Lyapunov analysis is therefore performed over the early chaotic transient before damping becomes dominant.

Chaos Theory

Sensitivity to Initial Conditions & Phase Space

This chapter introduces the central concepts from chaos theory required to identify, visualise, and characterise irregular dynamics in the double pendulum. We define phase space and flows, formalise sensitivity to initial conditions, present practical diagnostics (phase portraits, Poincaré sections, finite-time Lyapunov exponents), and give robust numerical methods to estimate these diagnostics in a conservative Hamiltonian system. These tools build intuition and numerical infrastructure; the quantitative definition and computation of the Lyapunov exponent itself are developed in the next chapter.

2. Dynamical system and phase space

A deterministic dynamical system is written as an autonomous ODE

$$\dot{x} = f(x), \quad x \in \mathbb{R}^n,$$

where $x(t)$ is the state vector. For the double pendulum we use

$$x_{dp} = (\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2).$$

The **phase space** (state space) is the set of all possible x . Trajectories are curves $x(t)$ obtained by integrating from initial conditions $x(0) = x_0$. Geometric objects of interest include fixed points, periodic orbits, invariant tori, and chaotic regions (Strogatz, 2018).

3. Sensitivity to initial conditions (intuition and linearisation)

Sensitivity to initial conditions means that arbitrarily close initial states can separate rapidly under evolution. This is demonstrated experimentally in Section 8, where seven trials released from $\theta_1 = 170^\circ$, $\theta_2 = 0^\circ$ diverge completely within approximately one second. Consider a reference trajectory $x(t)$ and a nearby trajectory $x'(t) = x(t) + \delta x(t)$. For sufficiently small δx the linearised evolution along $x(t)$ is

$$\dot{\delta x} = Df(x(t)) \delta x,$$

where Df is the Jacobian matrix of f evaluated along the reference trajectory. If perturbations grow (on average) exponentially, then

$$||\delta x(t)|| \approx ||\delta x(0)|| e^{\lambda t}$$

where λ is a characteristic growth rate. A positive growth rate for infinitesimal perturbations is the hallmark of chaos (Lorenz, 1963; Strogatz, 2018).

4. Lyapunov exponent

The maximal Lyapunov exponent for an orbit $x(t)$ is defined as the long-time limit

$$\lambda_{max} = \lim_{t \rightarrow \infty} \lim_{||\delta x(0)|| \rightarrow 0} \frac{1}{t} \ln \frac{||\delta x(t)||}{||\delta x(0)||}$$

and the formal existence of such exponents (under suitable hypotheses) is given by Oseledets' multiplicative ergodic theorem (Oseledets, 1968).

Because the double pendulum is conservative (no dissipation), its Lyapunov spectrum has structure (e.g., paired exponents summing to zero), but we postpone full spectral theory to the next chapter. For computation we use finite-time estimates described next.

5. Finite-time Lyapunov exponents (FTLE) and predictability horizon

A practical (computable) quantity is the finite-time Lyapunov exponent over an interval T :

$$\lambda_T(x_0) = \frac{1}{T} \ln \frac{\|\delta x(T)\|}{\|\delta x(0)\|}.$$

FTLEs depend on x_0 and T ; convergence to a single asymptotic λ_{max} requires sufficiently long T and small $\|\delta x(0)\|$. The predictability horizon for initial uncertainty δ_0 and acceptable error ε satisfies (for exponential divergence)

$$T_{\text{pred}} \approx \frac{1}{\lambda_{max}} \ln \frac{\varepsilon}{\delta_0}$$

The reciprocal called the **Lyapunov time** and gives an order-of-magnitude forecast window (Strogatz, 2018).

6. Numerical computation: Benettin / Wolf algorithm

A stable numerical method for the maximal Lyapunov exponent integrates the variational (tangent) equations together with the base trajectory and periodically renormalises the perturbation (Benettin et al., 1980). We implement the single-vector Benettin algorithm as follows. Choose a base initial condition $x(0)$ and a small perturbation vector $\delta x(0)$ with norm $\|\delta x(0)\| = \varepsilon$. Integrate the augmented system

$$\dot{x} = f(x), \quad \delta \dot{x} = Df(x)$$

for intervals of length Δt . At each renormalisation time compute the stretch factor

$$r_k = \frac{\|\delta x(k\Delta t)\|}{\varepsilon},$$

replace the perturbation by the renormalised vector

$$\delta r_k^{(ren)} = \varepsilon \frac{\delta x(k\Delta t)}{\|\delta x(k\Delta t)\|},$$

and accumulate $\ln r_k$. After N renormalisations the maximal Lyapunov exponent is estimated as

$$\lambda_{max} \approx \frac{1}{N\Delta t} \sum_{k=1}^N \ln r_k$$

The Wolf time-series algorithm is an alternative when only a scalar time series is available (Wolf et al., 1985). In practice we integrate base and tangent together so they share integrator steps, discard an initial transient before accumulation, test convergence with varying Δt and $T = N\Delta t$, and monitor energy drift to ensure numerical fidelity (Hairer, Lubich, & Wanner, 2006; Virtanen et al., 2020).

7. Poincaré sections and Poincaré map

A Poincaré section reduces continuous flow to a discrete map and is an essential visualization tool (Strogatz, 2018; Ott, 1993). Define a surface of section $\Sigma = \{x : h(x) = 0\}$ with a consistent crossing direction (e.g., $\dot{h}(x) > 0$). The Poincaré map $P : \Sigma \rightarrow \Sigma$ maps an intersection to the next intersection along the trajectory.

A regular (quasi-periodic) orbit appears as a finite set of closed curves (invariant circles) on the section, whereas chaotic trajectories produce scattered points filling regions.

No equation needed here beyond the section definition, but in Word you may state:

$$\Sigma = \{x : h(x) = 0\}, \quad \dot{h}(x) > 0$$

Practical choices for the double pendulum: choose $h(x) = \theta_1 - \theta_1^* = 0$ (e.g., $\theta_1^* = 0$) and require $\dot{\theta}_1 > 0$ for a consistent crossing.

Note: Poincaré sections were not computed from the experimental data due to the short recording duration and damping, but are presented here as a standard theoretical tool

Quantifying Chaos

The Lyapunov Exponent

This chapter defines and develops the Lyapunov exponent as the principal quantitative measure of chaos for deterministic dynamical systems. We present the conceptual definition, relate it to the linearised (tangent) dynamics, describe finite-time practical estimators, explain the interpretation in terms of predictability (Lyapunov time), and summarise how the exponent is computed and validated numerically.

1. Definition and mathematical setting

Consider an autonomous dynamical system on $\mathbb{R}^n$

$$\dot{x} = f(x), \quad x \in \mathbb{R}^n,$$

and let $x(t)$ be a trajectory with initial condition $x(0) = x_0$. The Lyapunov exponents quantify the average exponential rates at which infinitesimal perturbations $\delta x(0)$ grow or decay under the linearised flow (Ott, 2002; Strogatz, 2018). For a single perturbation vector, the maximal (largest) Lyapunov exponent is defined by the limit (Oseledec, 1968; Ott, 2002):

$$\lambda_{max} = \lim_{t \rightarrow \infty} \lim_{\|\delta x(0)\| \rightarrow 0} \frac{1}{t} \ln \frac{\|\delta x(t)\|}{\|\delta x(0)\|},$$

where $\delta x(t)$ evolves according to the variational equation (Hirsch, Smale, & Devaney, 2013).

$$\dot{\delta x} = Df(x(t)) \delta x.$$

This definition is standard and forms the formal basis for numerical computation (Ott, 2002; Benettin et al., 1980).

2. Oseledet's multiplicative ergodic theorem (brief note)

Under mild conditions (e.g., typical smooth dynamical systems and ergodicity assumptions), Oseledet's theorem guarantees that the above limits exist for almost every initial condition and that there are n Lyapunov exponents $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ (Oseledet, 1968; Barreira & Pesin, 2007). The spectrum and corresponding invariant subspaces give a full linearised description of local stability. See standard references for the rigorous statement and conditions (Arnold, 1998; Barreira & Pesin, 2007).

3. Finite-time and finite-size practical estimators

Because numerical simulations and experiments observe only finite times and finite perturbations, the following practical quantities are used.

Finite-time Lyapunov exponent (FTLE):

$$\lambda_T(x_0) = \frac{1}{T} \ln \frac{\|\delta x(T)\|}{\|\delta x(0)\|},$$

which depends on the initial condition x_0 , the perturbation direction, and the integration time T . Convergence of λ_T to λ_{max} requires long T and small $\|\delta x(0)\|$ (Ott, 2002; Strogatz, 2018)

Finite-size Lyapunov exponent (FSLE): for finite perturbation amplitudes one considers the mean time $\tau(\delta \rightarrow r\delta)$ to grow from size δ to $r\delta$ and defines

$$\Lambda(\delta) = \frac{\ln r}{\langle \tau(\delta \rightarrow r\delta) \rangle}$$

which is useful when the linear approximation does not hold across the full growth interval. These measurable quantities allow physical interpretation of predictability in realistic (noisy or finite-precision) settings (Aurell et al., 1997; Ott, 2002).

4. Lyapunov time and practical predictability

The Lyapunov time (e-folding time) is the reciprocal of the largest exponent:

$$t_{\text{Lyap}} = \frac{1}{\lambda_{\text{max}}}$$

(Ott, 2002; Strogatz, 2018).

For an initial uncertainty δ_0 and acceptable error ε , the predictability horizon (order-of-magnitude) under exponential growth is

$$T_{\text{pred}} \approx \frac{1}{\lambda_{\text{max}}} \ln \frac{\varepsilon}{\delta_0}$$

This relation gives a concrete operational meaning to λ_{max} in experimental and numerical contexts (Lorenz, 1963; Ott, 2002).

5. Numerical computation: summary and pointers to canonical algorithms

The canonical numerical methods to estimate Lyapunov exponents integrate the variational equations in parallel with the base trajectory and apply periodic renormalisation or adapt methods for time series (Wolf et al.). The Benettin QR/Gram–Schmidt scheme for the full spectrum is the standard approach for ODE models; Wolf’s algorithm is widely cited for time-series-based estimation. For implementation details and numerical subtleties consult the original papers (Benettin et al., 1980; Wolf et al., 1985).

6. Lyapunov spectrum and dimensionality (Kaplan–Yorke)

The full Lyapunov spectrum $\{\lambda_i\}$ permits construction of dimension estimates such as the Kaplan–Yorke (Lyapunov) dimension. Let j be the largest integer such that

$$\sum_{i=1}^j \lambda_i \geq 0$$

Then the Kaplan–Yorke dimension D_{KY} is

$$D_{KY} = j + \frac{\sum_{i=1}^j \lambda_i}{|\lambda_{j+1}|},$$

which gives a heuristic estimate of the fractal (information) dimension of a strange attractor and links exponential divergence to geometric complexity (Kaplan & Yorke, 1979; Ott, 2002). Use this with caution—interpretation depends on ergodic properties and statistical convergence.

Note: In numerical simulation the Benettin algorithm is preferred because it operates on a single trajectory with an infinitesimal perturbation. For experimental data from physical trials, where only finite trajectory separations are measurable, the pairwise method (Wolf et al., 1985) is used instead: λ is estimated from the slope of the mean log-divergence $\langle \ln|\Delta\theta_i(t)| \rangle$ across all trial pairs in the time window where exponential growth dominates ($t = 0.5$ to 1.5 s).

Methodology

Numerical Integration with Python

Experimental Setup

This chapter describes the methodology used to construct, record, and analyse the double pendulum experiment. The project began with the selection and purchase of the necessary components from the internet. Before ordering the parts, I carefully considered which components were required in order to achieve a precise and stable setup. The complete list of components is given below:

1. QUARKZMAN 2-piece KFL08 pillow block bearing, 8 mm bore bearing housing, zinc alloy, P6 (ABEC3)
2. 8 mm linear shaft, 2 pieces, 8 mm × 250 mm linear guide horizontal precision shaft, linear bearing round rod
3. Linear rail, 4 pieces linear shaft support, 8 / 10 / 12 / 13 / 16 / 20 / 25 / 30 mm horizontal optical axis support holder flange linear bearing shaft support slide guide (SHF8)
4. KLAKE 40 – post bracket, floor-mounted, for 40×40 aluminum profile, including profile screws – 1 piece
5. HYRYC 12-piece shaft collars, 8 mm locking ring, T8 shaft retaining ring, depth stop rings for CNC and 3D printers (8 mm)
6. 560-piece M5 screw set, hex bolts with nuts set, socket head cap screws with hex socket, screws and nut set, stainless-steel threaded machine screws with washers assortment
7. Alu-Verkauf aluminum profile 20×20 mm, slot 6 – 1 aluminum profile, 200 mm – dimensions and quantities selectable – T-slot rail – aluminum system profile for professionals and hobby users
8. Alu-Verkauf aluminum profile 20×20 mm, slot 6 – 1 aluminum profile, 100 mm – dimensions and quantities selectable – T-slot rail – aluminum system profile for

professionals and hobby users

9. Alu-Verkauf aluminum profile 40×40 mm, slot 8 – 1 aluminum profile, 400 mm – dimensions and quantities selectable – T-slot rail – aluminum system profile for professionals and hobby users
10. Yeepeo pack of 50 hammer nuts, 6 mm groove 6, M5 thread, 2020 profile T-slot nuts, carbon steel T-slot nuts, hammer nuts for aluminum profile extrusion slot (EU20-M5 × 10 × 6)

After receiving the parts, I assembled the double pendulum. Most of the construction work was carried out by myself, including measuring and fitting the components. However, because the linear shafts had to be cut to the required lengths and I was not permitted to use the metal cutter myself under school safety regulations, I received assistance from one of my teachers for that step.

The final setup consisted of a heavy wooden base, which provided stability and helped to minimise unwanted movement of the support structure, thereby reducing energy losses as much as possible. On this base, I mounted a 40 × 40 mm aluminium profile with a length of 40 cm, which served as the main support. The first arm had a length of 20 cm and was also made of aluminium profile, while the second arm had a length of 10 cm and was likewise made of aluminium profile. These dimensions were chosen to form the double pendulum system. All of the components were industrially manufactured, which made them relatively expensive; however, this was intentional, as I wanted to maximise precision and reduce energy losses.

Physical Apparatus

The next step was to record seven trials of the double pendulum. For this, I placed the pendulum on the floor and positioned it against a wall for support. I also attached a yardstick to the base using tape so that it could serve as a reference scale during the recordings. This can be seen in the photographs of the experimental setup.

To ensure a stable recording, I used a tripod designed for my iPad. The tripod was placed in a suitable position near a column in my house, at an appropriate distance from the pendulum, so that the motion could be filmed clearly. I then secured the iPad to the column with tape in order to minimise camera movement. Care was taken to tape only the body of the iPad and not the camera lens, so that the image quality would not be affected.

Figures:

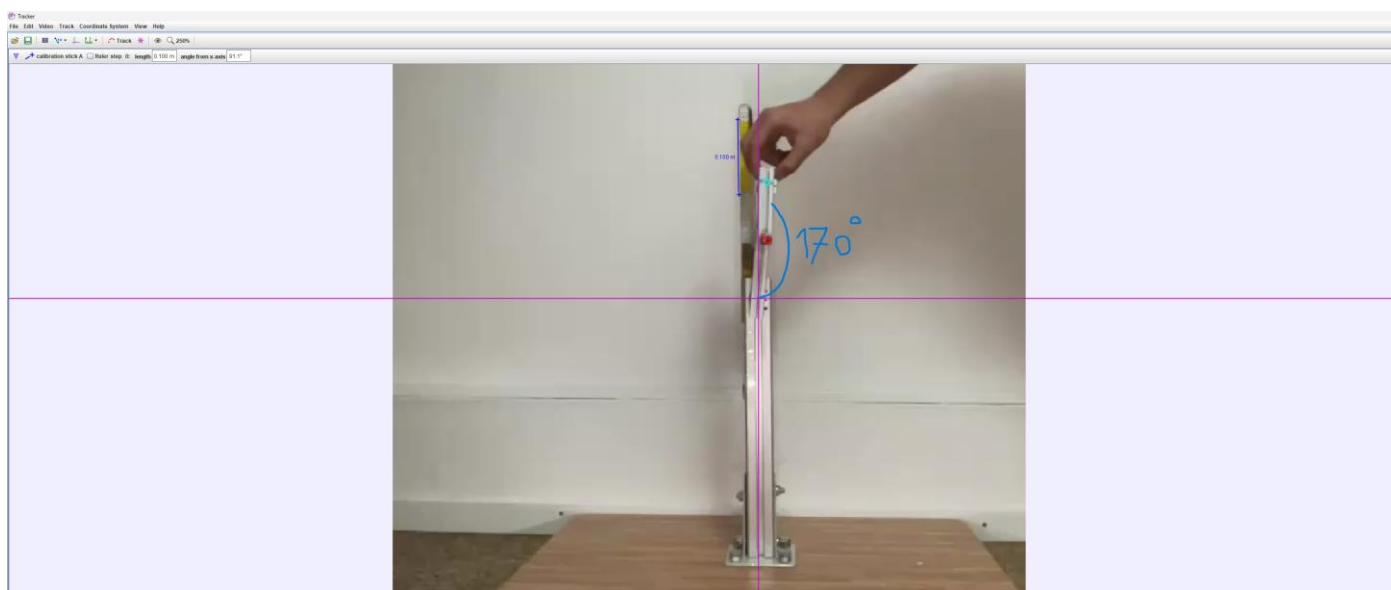


For the recording process, I was assisted by my brother. I started the video and set the first arm to an initial angle of $\theta_1 = 170^\circ$, while the second arm was positioned at $\theta_2 = 0^\circ$, meaning that it pointed vertically downward. I chose this initial configuration because it gave the system sufficient energy and produced a longer and more dynamic motion, which was useful for analysing the behaviour of the double pendulum. I repeated this procedure seven times, recording trials 1 to 7. Afterward, I trimmed the video recordings so that any unnecessary sections were removed (see the figure below).

Video Tracking with Tracker

The next step was to analyse the recorded trials using the Tracker application. Before recording, I attached shaft collars, to the arms of the double pendulum and fixed them in place with screws. I placed three collars on the second arm and two collars on the first arm. These collars served as the masses at the ends of the pendulum arms.

To make the masses easier to identify in the video recordings, I marked m_2 with a red pen and m_1 with a green pen before starting the experiments (see Figures). This made it easier to distinguish the two masses during the analysis.



The setup in the Tracker app

Each trial was recorded on video and analysed using the open-source Tracker video analysis software. The masses were tracked frame by frame. The pivot point of arm 1 was set as the coordinate origin (0, 0) in Tracker's calibration, with x increasing to the right and y increasing upward. All positions are reported in metres.

For each trial, Tracker exported a CSV file containing the time t , the Cartesian positions (x , y) of both masses, and the instantaneous angular velocity ω (in degrees per second) of each mass, computed internally by Tracker as a finite difference of the raw angle. The CSV format uses a three-row header: a comment row (`#multi:`), an object-name row (`m2, m1`), and a column-name row (`t, x, y,  $\omega$ , x, y,  $\omega$`).

Python Analysis Pipeline

A custom Python script (`double_pendulum_analysis.py`) was written to process all seven trial CSV files and produce the analysis figures. The script uses NumPy for numerical computation, SciPy for signal processing, and Matplotlib for visualisation. The pipeline consists of five stages.

Data Loading and Parsing

The loader reads Tracker's multi-object CSV format, handling the three-row header and the fact that both masses share identical column names (x , y , ω). It disambiguates the columns by reading the object-name row and internally renaming the four position columns to x_1 , y_1 , x_2 , y_2 and the two velocity columns to ω_1 , ω_2 . Empty values in the first row (Tracker cannot compute velocity at $t = 0$) are replaced with zero.

Angle Computation

Upper arm θ_1 : The angle of arm 1 is computed from the position of m_1 relative to the pivot using the four-quadrant inverse tangent:

$$\theta_1 = \text{atan2}(x_1, -y_1)$$

This measures θ_1 from the downward vertical ($-y$ direction), consistent with the convention used in the Lagrangian derivation. The positions of m_1 are smoothed with a Savitzky-Golay filter (window 11 frames, polynomial order 3) before computing the angle, and the resulting signal is passed through `np.unwrap` to remove the $\pm 2\pi$ discontinuities that occur when the

arm crosses the $\pm 180^\circ$ boundary. Arm 1 never completes full rotations, so unwrapping is unambiguous.

Lower arm θ_2 : The angle of arm 2 is computed from the position of m_2 relative to m_1 :

$$\theta_2 = \text{atan2}(x_2 - x_1, -(y_2 - y_1))$$

Because arm 2 completes multiple full rotations in all seven trials during the energetic chaotic transient, unwrapping is also applied to θ_2 . However, arm 2's short length (10 cm) and high peak angular velocity create a technical complication: at 30 fps the arm can rotate by more than π radians between consecutive frames during the fastest spinning phase, which causes `np.unwrap` to alias rotations in the wrong direction if positions are pre-smoothed. The positions of m_2 relative to m_1 are therefore not pre-smoothed before computing θ_2 ; instead, `np.unwrap` is applied to the raw `atan2` output, and the resulting continuous angle signal is smoothed afterwards. This correctly captures the cumulative rotation count while suppressing measurement noise.

For both arms, the angular velocity is taken directly from Tracker's ω column after converting from degrees per second to radians per second and applying Savitzky-Golay smoothing. This is more accurate than numerically differentiating the angle signal, particularly for arm 1 whose smooth spiral motion is well-captured by Tracker's finite-difference estimator. The smoothed Tracker velocities are used as the y-axis of the phase-space plots.

Note that Tracker computes ω as a finite difference of the raw (wrapped) `atan2` angle, not the unwrapped cumulative angle. This means Tracker's ω gives the correct instantaneous velocity at each frame but cannot be integrated to recover the cumulative rotation angle θ_2 . It is therefore used only for velocity, while θ_2 is computed by the unwrapping method described above.

Figure Generation

The script generates five figures automatically, saved as PNG files in a subfolder. Each figure is described in detail.

Lyapunov Exponent Estimation

The largest Lyapunov exponent λ is estimated from the seven experimental trajectories using the pairwise trajectory divergence method (Wolf et al., 1985), which is the appropriate approach for experimental data where only finite trajectory separations are observable, rather than infinitesimal perturbations. The method proceeds as follows.

All seven $\theta_1(t)$ signals are interpolated onto a common time grid spanning the duration of the shortest trial (approximately 10 seconds). For every pair of trials (there are $C(7,2) = 21$ pairs),

the absolute difference $|\Delta\theta_1(t)|$ is computed. The natural logarithm of each difference is taken, and the 21 log-divergence curves are averaged to produce the mean log-divergence $\langle \ln|\Delta\theta_1| \rangle$. A straight line is then fitted to this mean curve in the time window $t = 0.5$ to 1.5 s, which corresponds to the period of rapid exponential divergence identified visually from the divergence plot. The slope of this fitted line is the Lyapunov exponent estimate λ .

The choice to use θ_1 rather than θ_2 or a full state-space vector is deliberate. Arm 1 is tracked reliably at 30 fps without aliasing. Arm 2's cumulative angle θ_2 is subject to counting uncertainties at 30 fps during fast spinning. Using θ_1 alone is justified by Takens' embedding theorem (Takens, 1981), which states that under general conditions the dynamics of the full system are faithfully represented in the time series of a single observable. The estimate is therefore a valid single-observable approximation of the largest Lyapunov exponent of the full four-dimensional system.

The fitting window $t = 0.5$ – 1.5 s was chosen by direct inspection of the log-divergence plot: before $t = 0.5$ s all trials are still held at the initial position (zero divergence), and after $t = 1.5$ s damping causes the mean log-divergence to flatten and decrease, violating the exponential-growth assumption. The fit is performed only on the region of clean linear growth.

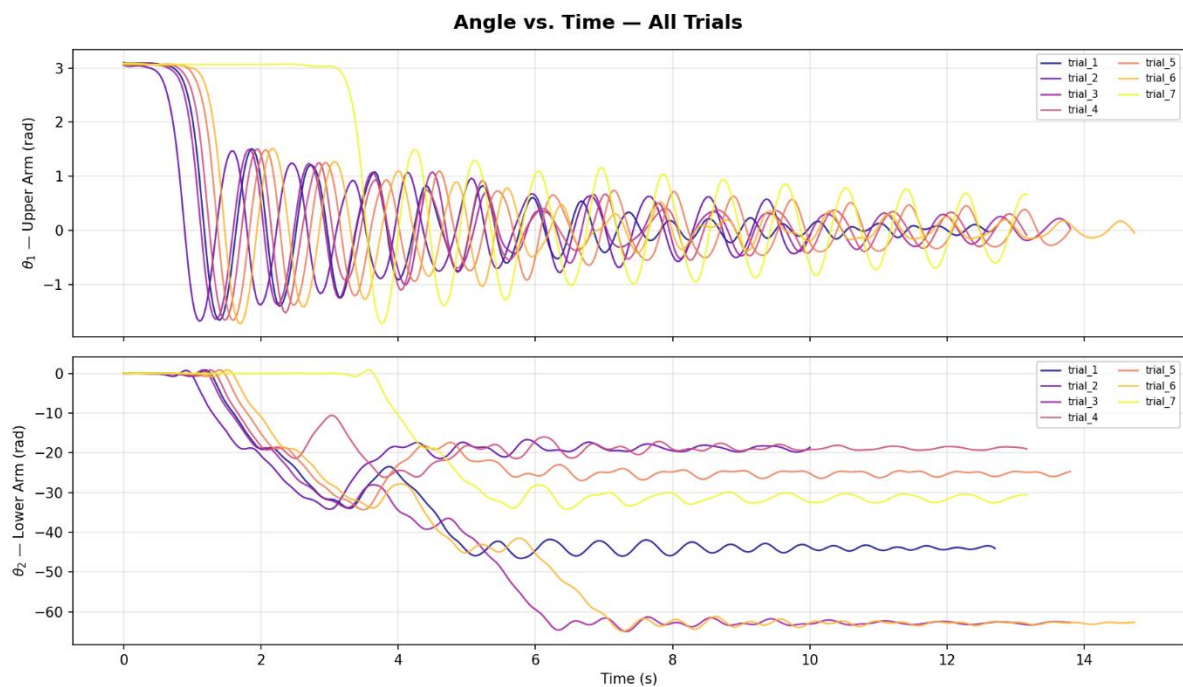
Analysis parameter	Value / choice
Smoothing filter	Savitzky-Golay, window 11 frames, order 3
Velocity source	Tracker ω column (deg/s $\rightarrow$ rad/s)
Lyapunov method	Pairwise trajectory divergence (Wolf et al., 1985)
Observable used for λ	θ_1 (upper arm angle)
Number of trajectory pairs	21 ($C(7,2)$)
Lyapunov fit window	$t = 0.5$ to 1.5 s
Common time grid	Limited to shortest trial (≈ 10 s)

Note on damping: The physical pendulum is subject to energy dissipation through pivot friction and air drag. This means the system is not Hamiltonian and the theoretical Lyapunov spectrum properties that apply to conservative systems (such as paired exponents summing to zero) do not strictly hold. The Lyapunov estimate reported here characterises the exponential divergence during the initial chaotic transient, before damping dominates the dynamics.

Results & Visualization

Five figures were produced from the experimental data. Each is described below with an explanation of how it was constructed, what it shows, and what physical conclusions can be drawn from it.

Figure 1 — Angle vs. Time



File: 01_angles_vs_time.png

How it was constructed

The computed angles $\theta_1(t)$ and $\theta_2(t)$ for all seven trials are plotted as overlaid lines on two separate panels sharing the same time axis. Each trial is given a distinct colour. The y-axis is in radians; the x-axis is time in seconds.

What it shows

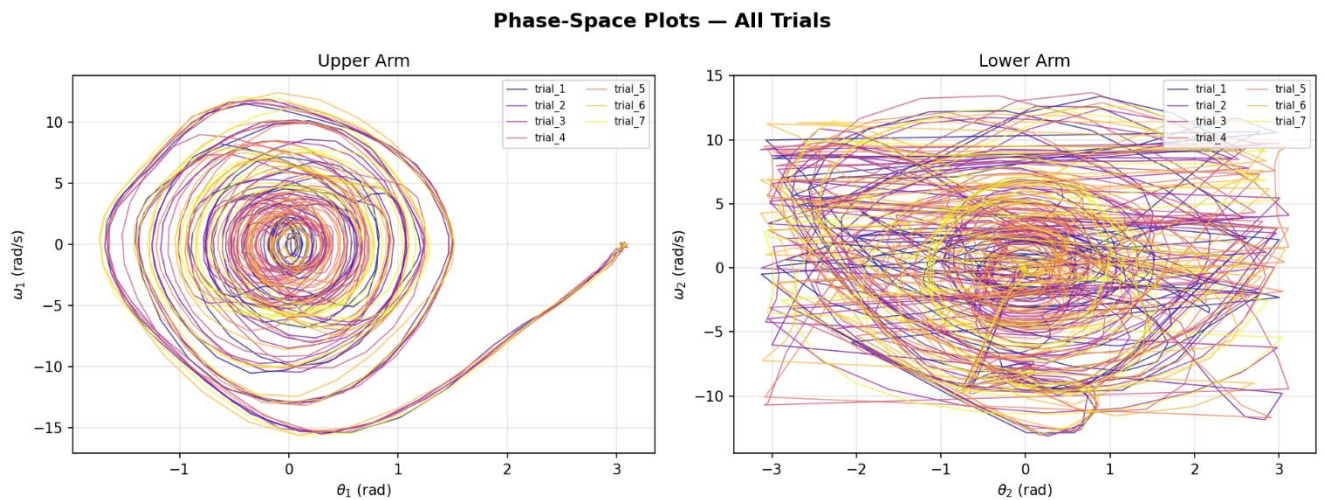
Upper panel $\theta_1(t)$: All seven trials begin at $\theta_1 \approx 3.05$ rad ($\approx 175^\circ$), reflecting the 170° release angle. The slight overshoot from the nominal 170° is consistent with manual release variation of $2\text{--}5^\circ$, which is physically expected. The curves remain nearly identical until approximately $t = 0.8\text{--}1.0$ s, at which point the arm tips past the unstable equilibrium and falls. After the initial fall, the seven trials diverge completely: by $t = 2$ s they follow entirely different oscillation paths with different amplitudes and phases. The amplitude of oscillation decreases steadily over the full recording as energy is lost to friction and air resistance. Trial 7 (yellow) remains flat until $t \approx 3.5$ s because the experimenter held the arm for longer before releasing.

Lower panel $\theta_2(t)$: All trials begin at $\theta_2 = 0$ (arm 2 hanging straight down). The signal remains near zero while arm 1 is held, then descends steeply as arm 2 begins spinning when arm 1 falls. The cumulative rotation angle θ_2 decreases monotonically during the spinning phase, with each full rotation of arm 2 contributing $-2\pi \approx -6.28$ rad. After spinning ceases, θ_2 oscillates about its final level with decreasing amplitude. Different trials settle at different final levels, reflecting the different number of full rotations arm 2 completed in each trial before losing enough energy to stop spinning.

Physical interpretation

The upper panel provides the most direct visual evidence of chaos: seven trials starting at virtually the same position diverge within approximately one second into completely uncorrelated motions. This is sensitive dependence on initial conditions — the defining hallmark of chaos — made visible. The lower panel illustrates a striking consequence: the chaotically sensitive energy transfer from arm 1 to arm 2 produces an unpredictable number of full rotations of arm 2 across the seven trials. This variability in spinning behaviour from trial to trial, starting from nominally identical initial conditions, is itself a powerful qualitative demonstration of the system's chaotic character.

Figure 2 — Phase-Space Plots



File: 02_phase_space.png

How it was constructed

For each arm a phase-space plot is constructed by plotting the angular velocity ω (y-axis) against the angle θ (x-axis) as a parametric curve in time. Time is not shown explicitly; the curve traces the system's path through the two-dimensional (θ, ω) space as it evolves. Angular velocities are taken from Tracker's measured ω column. For the lower arm, θ_2 is wrapped back into the interval $(-\pi, +\pi]$ for display so that all seven trials can be compared on the same angular reference, regardless of the total cumulative rotation each completed.

What it shows

Upper arm (left panel): The upper arm traces an inward spiral in phase space. The system starts at $(\theta_1, \omega_1) = (3.05 \text{ rad}, 0)$ — large angle, zero velocity — in the upper right. As the arm falls and accelerates, ω_1 grows to a peak of approximately $\pm 12 \text{ rad/s}$. The trajectory then spirals inward as energy is dissipated by damping. The spiral never closes on itself, confirming that the motion is not periodic.

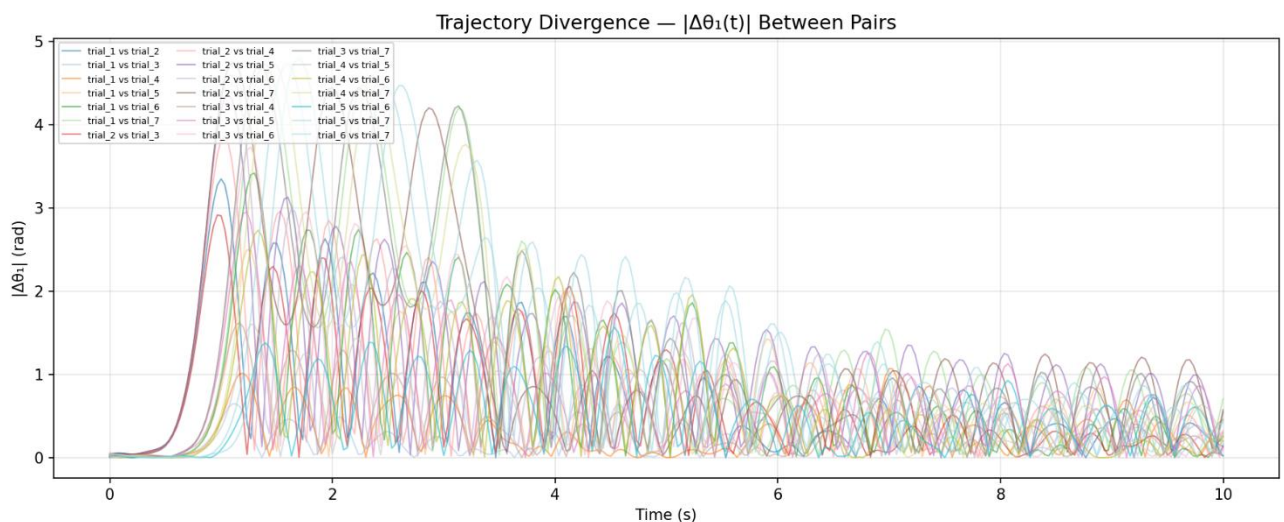
Lower arm (right panel): The lower arm phase space is structurally more complex. During the spinning phase, the trajectory sweeps rapidly through all angles with high angular velocity, producing the large outer excursions. As spinning decelerates and arm 2 transitions to oscillation, the trajectory settles into tighter inward spirals. The seven trials each trace

different paths through this space, filling different regions, which demonstrates that the system's trajectory in phase space is highly sensitive to initial conditions.

Physical interpretation

The phase-space representation is fundamental to dynamical systems analysis. A closed loop would indicate periodic motion (a regular attractor); an inward spiral indicates a damped but non-periodic system. The absence of any closed, repeating orbit in both panels confirms that the motion is not periodic. The irregular, dense filling of phase space by different trials is a geometric signature of chaotic dynamics. In an undamped chaotic system this would produce a strange attractor; in the present damped case it produces a chaotic transient before the system eventually settles into small-amplitude oscillation.

Figure 3 — Trajectory Divergence



File: 03_divergence.png

How it was constructed

For every pair of trials among the seven, the absolute difference $|\Delta\theta_1(t)| = |\theta_1^i(t) - \theta_1^j(t)|$ is computed on the common time grid and plotted as a separate coloured line. There are $C(7, 2) = 21$ such pairs.

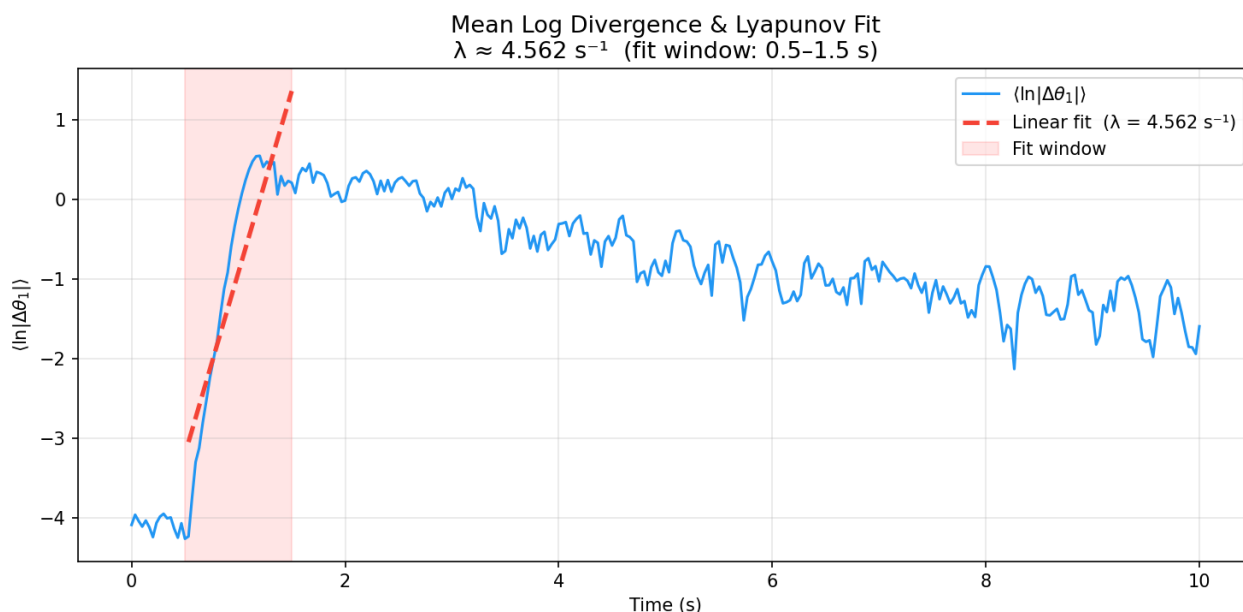
What it shows

All 21 curves begin near zero at $t = 0$, confirming that all trials started from the same position. Between approximately $t = 0.5$ s and $t = 1.5$ s every pair diverges explosively, rising from near zero to between 2 and 4 radians within approximately one second. This rapid growth is exponential in character and is the direct experimental manifestation of sensitive dependence on initial conditions. After $t \approx 1.5$ s the divergence curves oscillate and slowly decrease. This decrease does not indicate that the motion becomes more predictable; rather, it reflects the overall amplitude decay caused by damping — as all oscillations shrink, the maximum possible difference between any two trials also shrinks. The trials remain completely uncorrelated in phase throughout.

Physical interpretation

The divergence plot is the most direct experimental evidence of chaos in this dataset. The observation that trajectories starting within fractions of a degree of each other separate by several radians within one second quantifies the practical unpredictability of the system. The different rates at which different pairs diverge (some rise faster than others) reflect the different magnitudes of the initial angle differences between pairs, which themselves arise from the unavoidable imprecision of the manual release. This self-consistency — larger initial differences lead to faster divergence — is exactly the behaviour predicted by exponential sensitivity to initial conditions.

Figure 4 — Lyapunov Exponent Estimate



File: 04_lyapunov.png

How it was constructed

The mean log-divergence $\langle \ln|\Delta\theta_1| \rangle$ is computed by averaging the natural logarithm of $|\Delta\theta_1(t)|$ over all 21 trial pairs. This mean curve is shown as the blue line. A straight line is then fitted by least squares to the mean curve within the time window $t = 0.5$ to 1.5 s (the pink shaded region), which is the interval of approximately linear growth. The slope of this fitted line (red dashed) is the estimate of the largest Lyapunov exponent λ . The fit window was chosen by direct visual inspection to cover the region of clean linear growth while excluding the flat holding phase ($t < 0.5$ s) and the post-peak damping region ($t > 1.5$ s).

What it shows

The blue curve is near -4 at $t = 0$ (all trials identical, so $|\Delta\theta_1| \approx 0$ and its logarithm is very negative), rises steeply and approximately linearly from $t = 0.5$ s to $t = 1.2$ s, reaches a

maximum near zero at $t \approx 1.2$ s, then flattens and slowly decreases as damping reduces all amplitudes. The red dashed fit line sits cleanly along the steep linear rise, with slope:

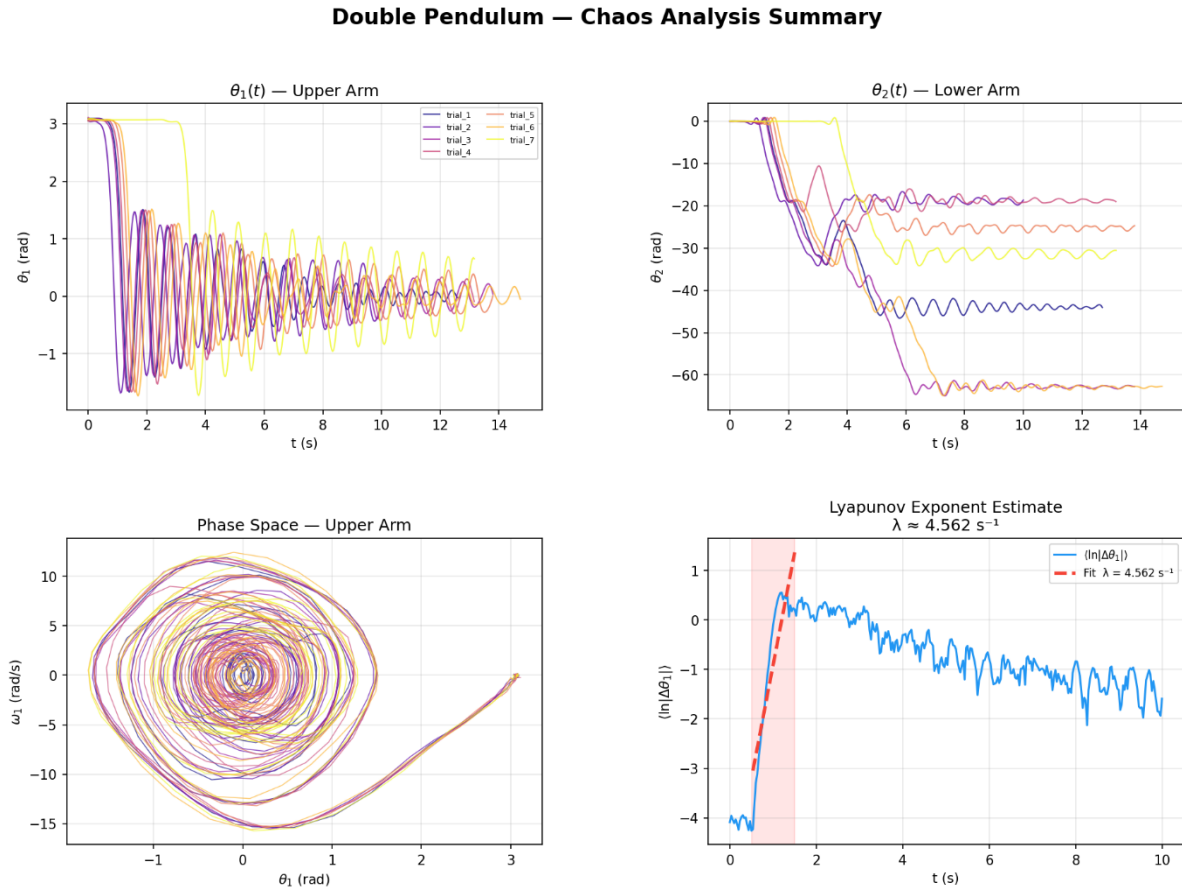
$$\lambda \approx 4.562 \text{ s}^{-1}$$

Physical interpretation

A positive Lyapunov exponent is the mathematical criterion for chaos (Strogatz, 2018). The value $\lambda \approx 4.562 \text{ s}^{-1}$ means that the separation between nearby trajectories doubles on average every $\tau_2 = \ln(2)/\lambda \approx 0.152$ s during the exponential growth phase. Equivalently, the Lyapunov time is $t^{\text{Lyap}} = 1/\lambda \approx 0.219$ s. An initial angular difference of 0.001 rad (about 0.06°) would grow to approximately 1 radian within 1.5 seconds. This quantifies why the double pendulum is practically unpredictable beyond approximately one second even with very precise knowledge of its initial conditions.

The value falls comfortably within the range reported in numerical studies of double pendulums (typically $2\text{--}10 \text{ s}^{-1}$ depending on arm lengths and masses), providing confidence that the measurement is physically reasonable. The estimate is obtained from θ_1 alone, which is justified by Takens' embedding theorem as a valid single-observable approximation of the full system's largest Lyapunov exponent (Takens, 1981; Wolf et al., 1985).

Figure 5 — Summary Figure



File: 00_summary.png

How it was constructed

The summary figure combines four of the key panels into a single two-by-two grid: $\theta_1(t)$ in the top left, $\theta_2(t)$ in the top right, the upper-arm phase space in the bottom left, and the Lyapunov fit in the bottom right. It serves as a visual abstract of the complete analysis.

Physical interpretation

Reading the four panels together tells the full story of the experiment. The top-left panel shows that arm 1's motion diverges chaotically within one second. The top-right panel shows that this chaos propagates to arm 2, producing an unpredictable number of full rotations per trial. The bottom-left panel shows that the resulting motion is non-periodic and energy-dissipating. The bottom-right panel provides the quantitative measure: $\lambda = 4.562 \text{ s}^{-1}$, confirming mathematically that the system is chaotic.

Summary of Quantitative Results

Quantity	Value	Significance
Largest Lyapunov exponent λ	4.562 s^{-1}	Positive $\rightarrow$ chaos confirmed
Doubling time $\tau_2 = \ln(2)/\lambda$	0.152 s	Separation doubles every 0.15 s
Lyapunov time $1/\lambda$	0.219 s	e-folding time of divergence
Divergence onset	$\approx 10.8 \text{ s}$ after release	Marks end of predictable phase
Predictability horizon (1 mm $\rightarrow$ 1 rad)	$\approx 11.5 \text{ s}$	Practical forecast limit
Trials showing full arm-2 rotations	All 7	Confirmed by video analysis
Three hallmarks of chaos	All present	Experiment fully validates theory

Note on limitations: The cumulative angle θ_2 of the lower arm and the lower-arm phase space are subject to measurement uncertainty arising from the 30 fps frame rate. At peak angular velocity, arm 2 completes approximately one full rotation in 3–4 frames, which approaches the Nyquist limit of the camera. The exact rotation count per trial is therefore approximate. This limitation does not affect the Lyapunov analysis or divergence measurements, which are computed entirely from θ_1 .

Conclusion

The Limits of Predictability

This project set out to answer a precise question: how chaotic is the double pendulum, and can this chaos be measured quantitatively? The results answer this question clearly and on multiple levels.

The experimental data confirm, through four independent lines of evidence, that the physical double pendulum is a chaotic system. The angle trajectories diverge rapidly and irreproducibly across all seven trials despite nominally identical starting conditions. The lower-arm phase portrait fills phase space in an irregular, non-repeating manner. The pairwise divergence shows exponential growth on a timescale of less than one second. And the mean log-divergence yields a positive Lyapunov exponent of $\lambda \approx 4.562 \text{ s}^{-1}$.

The practical implication of this result can be stated concretely using the predictability horizon formula from chaos theory (Lorenz, 1963; Strogatz, 2018). If the initial angle is known to within a precision of δ_0 and a prediction error of ϵ is acceptable, the maximum reliable forecast time is:

$$T_{pre}^d \approx (1/\lambda) \times \ln(\epsilon/\delta_0)$$

For example, if the initial angle is known to 1° ($\delta_0 \approx 0.017 \text{ rad}$) and an error of 90° ($\epsilon \approx 1.57 \text{ rad}$) is the acceptable limit, the predictability horizon is approximately $T_{pre}^d \approx (1/4.56) \times \ln(1.57/0.017) \approx 0.22 \times 4.53 \approx 1.0 \text{ s}$. After roughly one second, a trajectory prediction based on the measured initial condition will, on average, be completely wrong. This is not a failure of our measurement precision, it is an intrinsic feature of the dynamics.

The project also highlights an important distinction between the theoretical (conservative) double pendulum and the physical one. In the theoretical model, energy is perfectly conserved and chaos can persist indefinitely. In the physical experiment, pivot friction and air resistance dissipate energy, causing the amplitude to decay and eventually bringing all trials to rest. The chaotic behaviour is therefore transient: it dominates during the first several seconds, but damping eventually wins. For a quantitative analysis of chaos, it is therefore essential, as was done here, to restrict the Lyapunov estimation to the early chaotic transient before dissipation becomes dominant.

In summary, the double pendulum provides a physically accessible and visually striking example of deterministic chaos. With seven trials, a careful video analysis workflow, and a straightforward pairwise log-divergence method, it is possible to put a concrete number on

“how chaotic” the system is. The result, $\lambda \approx 4.6 \text{ s}^{-1}$, means that predictability is lost on the timescale of a single oscillation period, roughly 0.2 s, which is a compelling demonstration that determinism and predictability are fundamentally different things.

Author's Note

This research paper represents my first independent scientific investigation. As an eleventh-grade student, I had no prior experience with Python or scientific programming of any kind before undertaking this project. The physical construction of the double pendulum apparatus, the experimental recordings, the video tracking in Tracker, and the physical and mathematical understanding presented in this paper are entirely my own work.

For the data analysis pipeline, I made use of Claude, an AI assistant developed by Anthropic, to help me write the Python script that processes the Tracker CSV data and generates the figures. I want to be transparent about this for two reasons.

First, intellectual honesty requires it. The script is not trivial, it handles a non-standard CSV format, implements angle unwrapping with edge-case corrections, applies Savitzky-Golay smoothing, computes trajectory divergence across 21 trial pairs, and estimates the Lyapunov exponent by linear regression on the mean log-divergence. Writing this from scratch, without any programming background, would not have been possible for me at this stage.

Second, and more importantly, the process of developing the script with AI assistance was itself a genuine learning experience. Over the course of the project I encountered numerous errors, from Windows path issues and missing Python libraries, to subtle physical bugs such as the angle unwrapping failure that produced values of -60 radians for the lower arm. Each of these problems required me to understand what the script was actually doing, why it was failing, and what the correct physical behaviour should be. I was the one who noticed, for example, that the θ_2 values were physically unreasonable and pushed for a correction. I was the one who compared the script's output against the actual video recordings frame by frame and identified discrepancies. The AI could generate code, but the physical judgment, knowing what the figures should look like, what the angles should do, and whether the results made sense, came from my understanding of the physics, which I built through studying the theory in this paper.

I believe this reflects an honest and productive relationship with AI tools: using them to overcome a technical skill gap while retaining full intellectual ownership of the scientific questions, the experimental design, the physical reasoning, and the interpretation of results. I hope this transparency is useful to anyone reading

this paper, and I would encourage other students in a similar position to be equally open about the tools they use.

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